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**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**PHYSICAL SCIENCES: CHEMISTRY (P2)
FISIESE WETENSKAPPE: CHEMIE (V2)**

MAY/JUNE 2025/MEI/JUNIE 2025

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 22 pages.
Hierdie nasienriglyne bestaan uit 22 bladsye.**

QUESTION 1/VRAAG 1

- 1.1 B ✓✓ ACCEPT/AANVAAR 2 (2)
- 1.2 D ✓✓ (2)
- 1.3 D ✓✓ (2)
- 1.4 A ✓✓ (2)
- 1.5 B ✓✓ (2)
- 1.6 C ✓✓ (2)
- 1.7 D ✓✓ (2)
- 1.8 B ✓✓ (2)
- 1.9 C ✓✓ (2)
- 1.10 C ✓✓ (2)
- [20]**

QUESTION 2/VRAAG 2

- 2.1 Compounds with one or more multiple bonds between C atoms in the hydrocarbon chain. ✓✓ (2 or 0)
Verbindings met een of meer meervoudige bindings tussen C-atome in die koolwaterstofkettings. (2 of 0)

OR/OF

A hydrocarbon with two or more bonds between the C-atoms.
'n Koolwaterstof met twee of meer bindings tussen die C-atome.

OR/OF

Hydrocarbons containing not only single bonds between C atoms.
Koolwaterstowwe wat nie slegs enkelbindings tussen die C-atome bevat nie.

ACCEPT/AANVAAR:

Compounds with one or more double/triple bonds between C atoms in the hydrocarbon chain.
Verbindings met een of meer dubbel/trippelbindings tussen C-atome in die koolwaterstofkettings.

- 2.2
- 2.2.1 E ✓ (1)
- 2.2.2 F ✓ (1)
- 2.3 Ketones/Ketone ✓
 Aldehydes/Aldehiede ✓ (2)

2.4 Tertiary/Tersiêre ✓

The hydroxyl group/functional group (-OH) is bonded to a C atom that is bonded to three other C atoms. ✓

Die hidroksiel/funksionele groep (-OH) is gebind aan 'n C-atoom wat aan drie ander C-atome gebind is.

OR/OF

The functional group ($\begin{array}{c} | \\ -\text{C}- \\ | \\ \text{OH} \end{array}$) is bonded to three other C atoms.

Die funksionele groep ($\begin{array}{c} | \\ -\text{C}- \\ | \\ \text{OH} \end{array}$) is gebind aan drie ander C-atome.

(2)

2.5

2.5.1

Marking criteria:

- Correct stem, i.e. hexane. ✓
- Both substituent (ethyl and iodo) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- *Korrekte stam, d.i. heksaan.* ✓
- *Beide substituent (etiel en jodo) korrek geïdentifiseer.* ✓
- *IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.* ✓

3-ethyl-4-iodohexane/3-etiel-4-jodoheksaan ✓✓✓

(3)

2.5.2

Marking criteria/Nasienkriteria:

- Correct stem and substituents: methyl and propanol ✓
Korrekte stam en substituent: metiel en propanol
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓
IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.

2-methylpropan-1-ol/ 2-methyl-1-propanol/ methylpropan-1-ol/
methyl-1-propanol ✓✓
*2-metielpropan-1-ol/ 2-metiel-1-propanol / metielpropan-1-ol/
metiel-1-propanol*

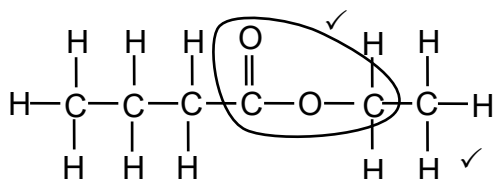
(2)

2.6

2.6.1 Esterification/Condensation/Verestering/Esterifikasie/Kondensasie ✓

(1)

2.6.2



Marking criteria/Nasienkriteria:

- Functional group correct. ✓
Funksionele groep korrek.
- Whole structure correct. ✓
Hele struktuur korrek.

IF/INDIEN

- More than one functional group/wrong functional group:
Meer as een funksionele groep/foutiewe funksionele groep: $0/2$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:* Max./Maks. $1/2$

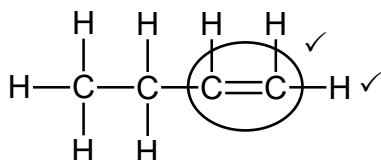
(2)

2.7

2.7.1 C_2H_4O ✓

(1)

2.7.2



Marking criteria/Nasienkriteria:

- Correct functional group. ✓
Korrekte funksionele groep.
- Whole structure correct. ✓
Hele struktuur korrek.

IF/INDIEN

- More than one functional group/wrong functional group:
Meer as een funksionele groep/foutiewe funksionele groep: $0/2$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:* Max./Maks. $1/2$

(2)

[19]

QUESTION 3/VRAAG 3

- 3.1.1 (A series of organic) compounds that can be described by the same general formula. ✓✓ (2 or 0)

OR

(A series of organic) compounds in which one member differs from the next by a CH_2 group.

(’n Reeks organiese) verbindings wat deur dieselfde algemene formule beskryf kan word. ✓✓ (2 of 0)

OF

(’n Reeks organiese) verbindings waarin die een lid van die volgende verskil met ’n CH_2 -groep

(2)

3.1.2

- (a) Formyl (group)/Formiel(groep) ✓

(1)

(b)

Marking criteria:

- Correct chain length, i.e. Meth. ✓
- Everything else correct. ✓

Nasienkriteria:

- Korrekte kettinglengte d.i. Met. ✓
- Alles verder reg ✓

Methanal/Metanaal ✓✓

(2)

3.1.3

- (a) Homologous series/Functional group/Type of intermolecular forces/Straight chain/Atmospheric pressure ✓

Homoloë reeks/Funksionele groep/Tipe intermolekulêre kragte/ Reguitketting/ Atmosferiese druk

(1)

- (b) The boiling points of the carboxylic acids increase with an increase in the chain length/the number of carbon atoms/surface area/molecular mass./

OR

The boiling points of the carboxylic acids decrease with a decrease in the chain length/number of carbon atoms/surface area/molecular mass. ✓

Die kookpunte van die karboksielsure neem toe met ’n toename in die kettinglengte/aantal koolstofatome/reaksieoppervlak/molekulêre massa./

OF

Die kookpunte van die karboksielsure neem af met ’n afname in die kettinglengte/aantal koolstofatome/reaksieoppervlak/molekulêre massa.

(1)

(c)

Marking criteria:

For increasing or decreasing number of C atoms

- Compare the strength of intermolecular forces. ✓
- Compare the energy required to overcome intermolecular forces. ✓

Nasienkriteria:*Vir toename of afname in aantal C-atome*

- *Vergelyk die sterkte van intermolekulêre kragte.* ✓
- *Vergelyk die energie benodig om intermolekulêre kragte te oorkom.* ✓

As the number of C atoms/chain length/surface area/contact area/molecular mass increases

- The strength of intermolecular/London/dispersion forces increases. ✓
- More energy is needed to overcome intermolecular forces/London/dispersion forces. ✓

OR

As the number of C atoms/chain length/surface area/contact area/molecular mass decreases

- The strength of intermolecular/London/dispersion forces decreases. ✓
- Less energy is needed to overcome intermolecular forces/London/dispersion forces. ✓

Met toename in aantal C-atome/kettinglengte/reaksieoppervlak/kontakarea/molekulêre massa.

- *Die sterkte van die die intermolekulêre kragte/Londonkragte/dispersiekragte neem toe.*
- *Meer energie word benodig om die intermolekulêre kragte/Londonkragte/dispersiekragte te oorkom/breek.*

OF*Met afname in aantal C-atome/kettinglengte/reaksieoppervlak/kontakarea/molekulêre massa.*

- *Die sterkte van die die intermolekulêre kragte/Londonkragte/dispersiekragte neem af.*
- *Minder energie word benodig om die intermolekulêre kragte/Londonkragte/dispersiekragte te oorkom/breek*

(2)

3.1.4 75 °C ✓

(1)

3.2

Marking criteria:

- Higher than ✓
- State that carboxylic acids have more than one (two) site for hydrogen bonding and alcohols have one site for hydrogen bonding. ✓
- Comparing the strength of IMFs. ✓
- Comparing the number of molecules in a vapour phase at a given temperature/energy needed to overcome IMFs. ✓

Nasienkriteria:

- Hoër as ✓
- Stel dat karboksiesure het meer as een (twee) plekke vir waterstofbindings en dat alkohole een plek het vir waterstofbinding. ✓
- Vergelyk die sterkte van die IMK's/energie benodig om IMK's te oorkom. ✓
- Vergelyk die hoeveelheid molekules in die dampfase by 'n gegewe temperatuur /energie nodig om die IMK te oorkom. ✓

- Higher than ✓
- Compound B/CH₃CH₂CH₂COOH/Carboxylic acid/Butanoic acid has (in addition to London forces and dipole-dipole forces), more than one site (two) for hydrogen bonding between molecules and compound A/CH₃CH₂CH₂CH₂CH₂OH/Alcohol/Pentan-1-ol has (in addition to London forces and dipole-dipole forces) one site for hydrogen bonding between molecules. ✓
- Intermolecular forces in compound B/CH₃CH₂CH₂COOH/Carboxylic acids/Butanoic acid are stronger. ✓
- More energy needed to overcome/break intermolecular forces in compound B/ CH₃CH₂CH₂COOH/Carboxylic acid/Butanoic acid.

OR

- At a given temperature there will be fewer molecules of compound B/CH₃CH₂CH₂COOH/Carboxylic acids/Butanoic acid in the vapour phase. ✓

- Higher than ✓
- Compound A/CH₃CH₂CH₂CH₂CH₂OH/Alcohol/Pentan-1-ol has (in addition to London forces and dipole-dipole forces) one site for hydrogen bonding between molecules and compound B/CH₃CH₂CH₂COOH/Carboxylic acid/Butanoic acid has, (in addition to London forces and dipole-dipole forces), more than one site (two) for hydrogen bonding between molecules. ✓
- Intermolecular forces in compound A/CH₃CH₂CH₂CH₂CH₂OH/Alcohol/ Pentan-1-ol are weaker. ✓
- Less energy needed to overcome/break intermolecular forces in compound A/CH₃CH₂CH₂CH₂CH₂OH/Pentan-1-ol/Alcohol.

OR

- At a given temperature there will be more molecules of compound A/CH₃CH₂CH₂CH₂CH₂OH/Alcohol/ Pentan-1-ol in the vapour phase. ✓

- Hoër as
 - Verbinding B/CH₃CH₂CH₂COOH/Karboksielsure/Butanoësuur het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), meer as een posisie (twee) vir waterstofbinding tussen molekule en verbinding A/CH₃CH₂CH₂CH₂CH₂OH/Alkohol/Pentan-1-ol het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), een posisie vir waterstofbinding tussen molekule.
 - Intermolekulêre kragte in verbinding B/CH₃CH₂CH₂COOH/Karboksielsure/Butanoësuur is sterker.
 - Meer energie word benodig om intermolekulêre kragte in verbinding B/CH₃CH₂CH₂COOH/Karboksielsure/Butanoësuur te oorkom/breek
- OF**
- By 'n gegewe temperatuur sal daar minder molekules van verbinding B/CH₃CH₂CH₂COOH/Karboksielsure/Butanoësuur in die dampfase wees.
-
- Hoër as
 - Verbinding A/CH₃CH₂CH₂CH₂CH₂OH/Alkohol/Pentan-1-ol het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), een posisie vir waterstofbinding tussen molekule en verbinding B/CH₃CH₂CH₂COOH/Karboksielsure/Butanoësuur het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), meer as een posisie (twee) vir waterstofbinding tussen molekule.
 - Intermolekulêre kragte in verbinding A/CH₃CH₂CH₂CH₂CH₂OH/Alkohol/Pentan-1-ol is swakker.
 - Minder energie word benodig om intermolekulêre kragte in A/CH₃CH₂CH₂CH₂CH₂OH/Alkohol/Pentan-1-ol te oorkom/breek.
- OF**
- By 'n gegewe temperatuur sal daar meer molekules van verbinding A/CH₃CH₂CH₂CH₂CH₂OH/Alkohol/Pentan-1-ol in die dampfase wees.

(4)
[14]

QUESTION 4/VRAAG 4

4.1

4.1.1 Hydrogenation/*Hidrogenering/Hidrogenasie* ✓ (1)4.1.2 Dehydration/*Dehidrasie/Dehidratering* ✓ (1)

4.2

Marking criteria:

- Correct chain length, i.e. But. ✓
- Everything else correct: IUPAC name completely correct including numbering. ✓

Nasienkriteria:

- *Korrekte kettinglengte d.i. But.* ✓
- *Alles verder reg: IUPAC-naam heeltemal korrek insluitende nommering.* ✓

Butan-1-ol/1-butanol ✓✓

(2)

4.3

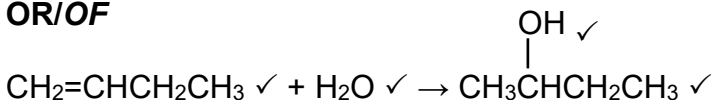
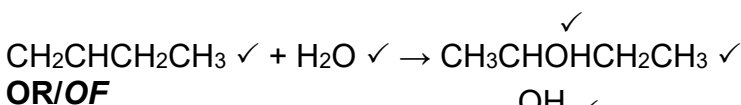
4.3.1

Marking criteria/Nasienkriteria:

- Whole condensed structural formula of alkene correct. ✓
Hele gekondenseerde struktuurformule van die alkeen korrek.
- H₂O. ✓
- Hydroxyl group/OH. ✓
Hidroksielgroep/OH.
- Whole condensed structural formula of alcohol correct (OH on second C-atom). ✓
Hele gekondenseerde struktuurformule van alkohol korrek (OH op tweede C-atoom)

IF/INDIEN

- Any additional reactants or products /*Enige addisionele reaktanse of produkte:*
Deduct 1 mark/*Trek 1 punt af.*
 - Structural formulae used/*Struktuurformule gebruik.* Max./Maks. $\frac{3}{4}$
 - Molecular formulae used/*Molekulêre formule gebruik.* Max./Maks. $\frac{1}{4}$
 - Only reactants without arrow/*Slegs reaktanse sonder pyl* Max/Maks. $\frac{2}{4}$
- Marking rule 6.3.10/*Nasienreël 6.3.10*



(4)

4.3.2 Sulphuric acid/H₂SO₄/Phosphoric acid/H₃PO₄/Swawelsuur/Fosforsuur ✓

(1)

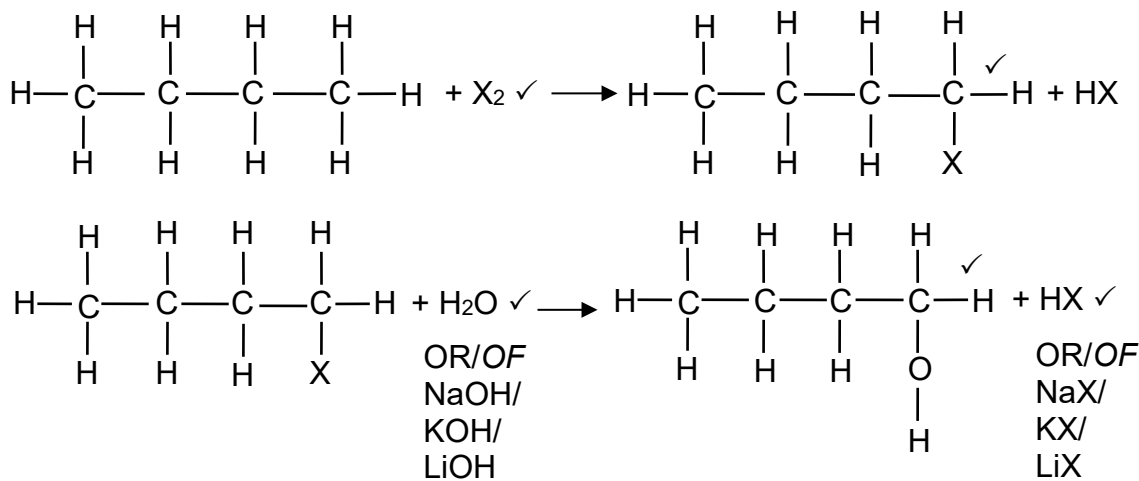
4.4

Marking criteria/Nasienkriteria:

- $X_2 = Br_2/Cl_2$. ✓
- Whole structural formula of haloalkane correct. ✓
Hele struktuurformule van haloalkaan korrek.
- $H_2O/NaOH/KOH/LiOH$. ✓
- Whole structural formula of alcohol correct. ✓
Hele struktuurformule van alkohol korrek.
- $HX/NaX/KX/LiX$ where/waar $X = Br/Cl$ ✓

IF/INDIEN

- Any additional reactants or products /Enige addisionele reaktanse of produkte:
Max./Maks. $\frac{4}{5}$
- Condensed structural formulae used/Gekondenseerde struktuurformule gebruik:
deduct 1 mark/trek 1 punt af.
- If inorganic product does not correspond with inorganic reactant: no mark for inorganic product./Indien anorganiese produk nie met die anorganiese reaktans ooreenstem nie, geen punt vir anorganiese produk.
- Molecular formulae used:/Molekulêre formule gebruik: Max./Maks. $\frac{3}{5}$
- Marking rule 6.3.10/Nasienreël 6.3.10



(5)

4.5



Ignore phases./Ignoreer fases.

Marking criteria/Nasienkriteria:

- C_4H_{10} ✓ O_2 , CO_2 and/en H_2O ✓ Balancing/Balansering ✓
- Ignore double arrows./Ignoreer dubbelpyle.
- Marking rule 6.3.10/Nasienreël 6.3.10.

IF/INDIEN:

- Structural formulae C_4H_{10} used:/Struktuurformule C_4H_{10} gebruik: Max./Maks. $\frac{2}{3}$
- Balancing mark only if everything else is correct/
Balanseringspunt slegs indien alles korrek.

(3)

[17]

QUESTION 5/VRAAG 5

5.1

NOTE/NOTAGive the mark for per unit time only if in context of reaction rate.Gee die punt vir per eenheidtyd slegs indien in konteks van reaksietempo.**ANY ONE:**

- Change in concentration ✓ of products/reactants per (unit) time. ✓
- Change in amount/number of moles/volume/mass of products or reactants per (unit) time.
- Amount/number of moles/volume/mass of products formed/reactants used per (unit) time.
- Rate of change in concentration/amount/number of moles/volume/mass. ✓✓ **(2 or 0)**

ENIGE EEN:

- Verandering in konsentrasie ✓ van produkte/reaktanse per (eenheid)tyd. ✓
- Verandering in hoeveelheid/getal mol/volume/massa van produkte of reaktanse per (eenheid)tyd.
- Hoeveelheid/getal mol/volume/massa van produkte gevorm/reaktanse gebruik per (eenheid)tyd.
- Tempo van verandering in konsentrasie/ hoeveelheid/getal mol/ volume/ massa. ✓✓ **(2 of 0)**

(2)

5.2

ANY ONE:

Temperature ✓/

(Initial) amount/Mass of magnesium carbonate/Surface area

ENIGE EEN:

Temperatuur ✓/

(Aanvanklike) hoeveelheid/Massa van magnesiumkarbonaat/

Reaksieoppervlak

(1)

5.3

CO₂/gas escapes from the reaction flask. ✓CO₂ /gas ontsnap uit die reaksiefles. ✓

(1)

5.4

Marking criteria: (a) Mass subtraction ✓ (b) Formula: $n = \frac{m}{M}$ or $V = nV_m$ ✓ (c) Substitute $M = 44 \text{ g} \cdot \text{mol}^{-1}$ in $n(\text{CO}_2) = \frac{m}{M}$ with $m(\text{CO}_2)$ from (a) ✓ (d) Substitute $24,5 \text{ dm}^3$ in $V = nV_m$ with $n(\text{CO}_2)$ ✓ (e) Substitute V_{CO_2} and 120 in rate formula ✓ (f) Final correct answer: $2,92 \times 10^{-3} (\text{dm}^3 \cdot \text{s}^{-1})$ ✓ Range: $2,08 \times 10^{-3}$ to 3×10^{-3}	$m(\text{CO}_2) = 144,5 - 143,87 \checkmark \text{ (a)}$ $= 0,63 \text{ g}$ $n(\text{CO}_2) = \frac{m}{M} \checkmark \text{ (b)}$ $= \frac{0,63}{44} \checkmark \text{ (c)}$ $= 1,43 \times 10^{-2} \text{ mol}$ $V(\text{CO}_2) = nV_m$ $= (1,43 \times 10^{-2})(24,5) \checkmark \text{ (d)}$ $= 0,35 \text{ dm}^3$ Ave rate/ gem tempo = $\frac{\Delta V(\text{CO}_2)}{\Delta t}$ $= \frac{0,35 - (0)}{120 - (0)} \checkmark \text{ (e)}$ $= 2,92 \times 10^{-3} (\text{dm}^3 \cdot \text{s}^{-1}) \checkmark \text{ (f)}$
Nasienkriteria: (a) Afrek van massas. ✓ (b) Formula: $n = \frac{m}{M}$ or $V = nV_m$ ✓ (c) Vervang $M = 44 \text{ g} \cdot \text{mol}^{-1}$ in $n(\text{CO}_2) = \frac{m}{M}$ met $m(\text{CO}_2)$ van (a) ✓ (d) Vervang $24,5 \text{ dm}^3$ in $V = nV_m$ met $n(\text{CO}_2)$ ✓ (e) Vervang V_{CO_2} en 120 in tempoformule ✓ (f) Finale korrekte antwoord: $2,92 \times 10^{-3} (\text{dm}^3 \cdot \text{s}^{-1})$ ✓ Gebied: $2,08 \times 10^{-3}$ tot 3×10^{-3}	

(6)

5.5

Marking criteria: • A ✓ • Comparison of the curves of the graph ✓ • Comparison of concentration of HCl (from table) ✓ • Explanation of collision theory for LOWER concentration ✓✓ Nasienkriteria: • A ✓ • Vergelyk die kurwes van grafiek ✓ • Vergelyk die konsentrasie van HCl (vanaf tabel) ✓ • Verduidelik botsingsteorie vir LAER konsentrasie ✓✓
--

A ✓

- Gradient is least steep/lowest reaction rate/least amount of gas produced in 120 s. ✓
- Lowest concentration of $\text{HCl}(\text{aq})$. ✓
- Least/Less particles per unit volume. ✓
- Least/Less effective collisions per unit time/second. ✓ **OR**
Lowest/Lower frequency of effective collisions.
- *Gradient is die laagste/laagste reaksietempo/minste hoeveelheid gas geproduseer in 120 s.* ✓
- Laagste konsentrasie van $\text{HCl}(\text{aq})$. ✓
- Minste/Minder deeltjies per eenheidsvolume. ✓
- Minste/Minder effektiewe botsings per eenheidstyd/sekonde. ✓ **OF**
Laagste/Laer frekwensie van effektiewe botsings.

(5)

5.6

The same/ *Dieselfde* ✓

The same amount MgCO_3 is used in each experiment. ✓

Dieselfde hoeveelheid MgCO_3 is gebruik in elke eksperiment.

(2)

[17]

QUESTION 6/VRAAG 6

6.1

6.1.1 Remains the same/*Bly dieselfde* ✓ (1)6.1.2 Decreases/*Neem af* ✓ (1)6.1.3 Remains the same/*Bly dieselfde* ✓ (1)

- 6.2
- Decrease in pressure favours the reaction that produces a greater number of moles/amount of gas. ✓
'n Verlaging in druk bevoordeel die reaksie wat 'n groter aantal mol/hoeveelheid gas produseer.
 - Forward reaction is favoured. ✓ / [CO] increases AND [CO₂] decreases
Voorwaartse reaksie word bevoordeel./ [CO] neem toe EN [CO₂] neem af (2)

6.3

Marking criteria: (a) Substitute 44 in $n = \frac{m}{M}$ ✓ (b) Change in mass of carbon: $m(C_i) - m(C_f)/n(C_i) - n(C_f)$ ✓ (c) Substitute 12 in $n = \frac{m}{M}$ ✓ (d) Use mole ratio 1:1 ✓ (e) $n(\text{CO}_2)_{\text{eq}} = n(\text{CO}_2)_{\text{initial}} - n(\text{CO}_2)_{\text{used}}$ OR $m(\text{CO}_2)_{\text{eq}} = m(\text{CO}_2)_{\text{initial}} - m(\text{CO}_2)_{\text{used}}$ ✓ (f) Final answer: 6,16 g ✓ RANGE: 6 to 6,16 g NOTE: If (b) $\Delta m(C)$ or $\Delta n(C)$ is not calculated max 2/6	Nasienkriteria: (a) Vervang 44 in $n = \frac{m}{M}$ ✓ (b) Verandering in massa: $m(C_i) - m(C_f)/n(C_i) - n(C_f)$ ✓ (c) Vervang 12 in $n = \frac{m}{M}$ ✓ (d) Gebruik molverhouding 1:1 ✓ (e) $n(\text{CO}_2)_{\text{ewe}} = n(\text{CO}_2)_{\text{begin}} - n(\text{CO}_2)_{\text{gebruik}}$ OF $m(\text{CO}_2)_{\text{ewe}} = m(\text{CO}_2)_{\text{begin}} - m(\text{CO}_2)_{\text{gebruik}}$ ✓ (f) Finale antwoord: 6,16 g ✓ GEBIED: 6 tot 6,16 g NOTA: Indien (b) $\Delta m(C)$ of $\Delta n(C)$ nie bereken maks 2/6
OPTION 1/OPSIE 1: $\Delta m(C) = 14 - 4,44$ ✓ (b) $= 9,56 \text{ g}$ $n(\text{CO}_2)_{\text{initially}} = \frac{m}{M}$ $= \frac{41,2}{44}$ ✓ (a) $= 0,94 \text{ mol (0,936)}$ $n(C)_{\text{used}} = \frac{m}{M}$ $= \frac{9,56}{12}$ ✓ (c) $= 0,80 \text{ mol (0,797)}$ $n(\text{CO}_2)_{\text{used}} = n(C)$ $= 0,80 \text{ mol (0,797)}$ ✓ (d) $n(\text{CO}_2)_{\text{eq}} = n(\text{CO}_2)_{\text{initially}} - n(\text{CO}_2)_{\text{used}}$ $= 0,94 - 0,80$ ✓ (e) $= 0,14 \text{ mol}$ $n(\text{CO}_2) = \frac{m}{M}$ $0,14 = \frac{m}{44}$ $X = m(\text{CO}_2) = 6,16 \text{ (g)}$ ✓ (f)	OPTION 2/OPSIE 2: $\Delta m(C) = 14 - 4,44$ ✓ (b) $= 9,56 \text{ g}$ $n(C)_{\text{used}} = \frac{m}{M}$ $n(C)_{\text{used}} = \frac{9,56}{12}$ ✓ (c) $= 0,80 \text{ mol (0,797)}$ $n(\text{CO}_2)_{\text{used}} = n(C)$ $= 0,80 \text{ mol (0,797)}$ ✓ (d) $n(\text{CO}_2) = \frac{m}{M}$ $0,80 = \frac{m}{44}$ ✓ (a) $m(\text{CO}_2) = 35,05 \text{ g}$ $m(\text{CO}_2)_{\text{eq}} = m(\text{CO}_2)_{\text{initially}} - m(\text{CO}_2)_{\text{used}}$ $= 41,2 - 35,05$ ✓ (e) $X = 6,15 \text{ (g)}$ ✓ (f)

OPTION 3/OPSIE 3:

$$\begin{aligned}
 n(\text{CO}_2)_{\text{initially}} &= \frac{m}{M} \\
 &= \frac{41,2}{44} \checkmark \text{ (a)} \\
 &= 0,94 \text{ mol (0,936)}
 \end{aligned}$$

$$\begin{aligned}
 n(\text{C})_{\text{used}} &= \frac{9,56}{12} \checkmark \text{ (c)} \\
 &= 0,80 \text{ mol (0,797)}
 \end{aligned}$$

	C	CO ₂
Ratio/ <i>Verhouding</i>	1	1
Initial quantity (mol) Aanvangshoeveelheid (mol)	1,17	0,936
Change (mol) <i>Verandering (mol)</i>	0,8	0,8 $\checkmark$ (d)
Quantity at equilibrium (mol)/ <i>Hoeveelheid by ewewig (mol)</i>	0,37 $\checkmark$ (b)	0,14 $\checkmark$ (e)

$$n(\text{CO}_2) = 0,139 \text{ mol}$$

$$m(\text{CO}_2) = 0,139 (44)$$

$$X = 6,16 \text{ (g)} \checkmark \text{ (f)}$$

(6)

6.4

POSITIVE MARKING FROM QUESTION 6.3:**POSITIEWE NASIEN VANAF VRAAG 6.3:****Marking criteria**

- (a) Use of ratio $n(\text{CO}_2) : n(\text{CO}) = 1:2$. ✓
 (b) Divide by 3 dm^3 . ✓
 (c) Correct K_c expression (formulae in square brackets). ✓
 (d) Substitute of concentration into K_c expression. ✓
 (e) Final answer: 5,98 ✓
 RANGE: 5,98 – 7,29

Nasienkriteria:

- (a) Gebruik verhouding $n(\text{CO}_2) : n(\text{CO}) = 1:2$. ✓
 (b) Deel deur 3 dm^3 . ✓
 (c) Korrekte K_c uitdrukking (formules in vierkantige hakies). ✓
 (d) Vervang konsentrasies in korrekte K_c uitdrukking. ✓
 (e) Finale antwoord: 5,98 ✓
 GEBIED: 5,98 – 7,29

NOTE/NOTA:

Mark calculations of this question that may be done in QUESTION 6.3.

Merk berekeninge van hierdie vraag wat in VRAAG 6.3 gedoen is.

CALCULATIONS USING NUMBER OF MOLES**BEREKENINGE WAT AANTAL MOL GEBRUIK****OPTION 1/OPSIE 1:**

$$n(\text{CO}_2)_{\text{initial}} = \frac{m}{M}$$

$$= \frac{41,2}{44}$$

$$= 0,936 \text{ mol}$$

	CO_2	CO
Ratio/Verhouding	1	2
Initial quantity (mol) Aanvangshoeveelheid (mol)	0,936	0
Change (mol) Verandering (mol)	0,8	1,6
Quantity at equilibrium (mol)/ Hoeveelheid by ewewig (mol)	0,14	1,6
Equilibrium concentration ($\text{mol} \cdot \text{dm}^{-3}$) Ewewigskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)	0,047	0,53

✓ (a)

Divide by/deel deur 3 ✓ (b)

$$K_c = \frac{[\text{CO}]^2}{[\text{CO}_2]} \quad \checkmark \text{ (c)}$$

$$= \frac{(0,53)^2}{0,047} \quad \checkmark \text{ (d)}$$

$$= 5,98 \quad \checkmark \text{ (e)}$$

Wrong K_c expressionVerkeerde K_c -uitdrukking: Max./Maks. $\frac{2}{5}$ No K_c expression/Geen K_c - uitdrukking: $\frac{4}{5}$

CALCULATIONS USING CONCENTRATION**BEREKENINGE WAT KONSENTRASIE GEBRUIK****OPTION 2/OPSIE 2:**

$$c(\text{CO}_2) = \frac{m}{MV}$$

$$= \frac{41,2}{(44)(3)}$$

$$= 0,31 \text{ mol} \cdot \text{dm}^{-3}$$

Divide by
3 ✓ (b)

$$c = \frac{n}{V}$$

$$= \frac{0,8}{3}$$

$$= 0,27 \text{ mol} \cdot \text{dm}^{-3} (0,267)$$

	CO ₂	CO
Ratio/Verhouding	1	2
Initial concentration (mol·dm ⁻³) Aanvangskonsentrasie (mol·dm ⁻³)	0,31	0
Change in concentration (mol·dm ⁻³) Verandering in konsentrasie (mol·dm ⁻³)	0,27	0,54
Equilibrium concentration (mol·dm ⁻³) Ewewigskonsentrasie (mol·dm ⁻³)	0,04	0,54

✓ (a)

$$K_c = \frac{[\text{CO}]^2}{[\text{CO}_2]} \quad \checkmark \text{ (c)}$$

$$= \frac{(0,54)^2}{0,04} \quad \checkmark \text{ (d)}$$

$$= 7,29 \quad \checkmark \text{ (e)}$$

Wrong K_c expressionVerkeerde K_c-uitdrukking: Max./Maks. 2/5No K_c expression/Geen K_c-uitdrukking: 4/5

OPTION 3/OPSIE 3:

$$n(\text{CO}_2)_{\text{initial}} = \frac{m}{M}$$

$$= \frac{41,2}{44}$$

$$= 0,936 \text{ mol}$$

$$\Delta n(\text{CO}_2) = 0,8 \text{ mol}$$

$$n(\text{CO}_2)_{\text{eqm}} = n(\text{CO}_2)_{\text{initial}} - \Delta n(\text{CO}_2)$$

$$= 0,936 - 0,8$$

$$= 0,136 \text{ mol}$$

$$n(\text{CO})_{\text{formed}} = 2\Delta n(\text{CO}_2)_{\text{used}} \checkmark \text{ (a)} = 1,6 \text{ mol}$$

$$n(\text{CO})_{\text{eqm}} = \Delta n(\text{CO})_{\text{formed}} = 1,6 \text{ mol}$$

$$\left. \begin{aligned} [\text{CO}_2]_{\text{eqm}} &= \frac{0,136}{3} = 4,53 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3} \\ [\text{CO}]_{\text{eqm}} &= \frac{1,6}{3} = 0,53 \text{ mol} \cdot \text{dm}^{-3} \end{aligned} \right\} \checkmark \text{ (b)}$$

$$K_c = \frac{[\text{CO}]^2}{[\text{CO}_2]} \checkmark \text{ (c)}$$

$$= \frac{(0,53)^2}{4,53 \times 10^{-2}} \checkmark \text{ (d)}$$

$$= 6,2 \checkmark \text{ (e)}$$

Wrong K_c expression

Verkeerde K_c -uitdrukking: Max./Maks. $\frac{2}{5}$

No K_c expression/Geen K_c - uitdrukking: $\frac{4}{5}$

(5)

6.5 Y ✓✓

(2)

6.6 Remains the same/Bly dieselfde ✓

(1)

[19]

QUESTION 7/VRAAG 7

7.1 An acid produces hydrogen ions /H⁺/hydronium ions/ H₃O⁺ in aqueous solution/water. ✓✓ (2 or 0)
 'n Suur is 'n stof wat waterstofione/H⁺/hidroniumione/H₃O⁺ vorm in waterige oplossing/water. (2 of 0) (2)

7.2
 7.2.1 (COOH)₂ ✓ (1)

7.2.2 NaCl ✓ (1)

7.2.3 HCO₃⁻ ✓
OR/OF NH₃ (1)

7.2.4 NaOH ✓✓
OR/OF Mg(OH)₂ (2)

7.3	Marking criteria: (a) Calculate n(H_xY) ✓ (b) Calculate n(NaOH) ✓ (c) Final answer: x = 2 ✓ (d) Reactants ✓ Products ✓ Balancing ✓ NOTE: Ignore ⇌ and phases Marking rule 6.3.10 OPTION 1/OPSIE 1: $n = cV$ $n_{\text{acid}} = (0,11)(0,02364) \checkmark \text{ (a)}$ $= 2,6 \times 10^{-3}$ $n_{\text{base}} = (0,26)(0,02) \checkmark \text{ (b)}$ $= 5,2 \times 10^{-3} \quad (0,0052)$ $\frac{n(\text{H}_x\text{Y})}{n(\text{NaOH})} = \frac{n_a}{n_b}$ $\frac{2,6 \times 10^{-3}}{5,2 \times 10^{-3}} = \frac{1}{n_b}$ $n_b = 2$ $\therefore x = 2 \checkmark \text{ (c)}$ <u>H₂Y(aq) + 2NaOH(aq) ✓ → Na₂Y (aq) + 2H₂O(l) ✓</u> Bal ✓ (d)	Nasienkriteria: (a) Bereken n(H_xY) ✓ (b) Bereken n(NaOH) ✓ (c) Finale antwoord: x = 2 ✓ (d) Reaktanse ✓ Produkte ✓ Balansering ✓ NOTA: Ignoreer ⇌ en fases Nasienreël 6.3.10 OPTION 2/OPSIE 2: $\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$ (a) ✓ $\frac{(23,64)(0,11)}{(20)(0,26)} = \frac{1}{n_b}$ (b) ✓ $n_b = 2$ $\therefore x = 2 \checkmark \text{ (c)}$ H_xY : NaOH $2,6 \times 10^{-3} : 5,2 \times 10^{-3}$ $1 : 2$
-----	--	---

(6)

7.4

Marking criteria:

- (a) Any formula: $\text{pH} = -\log[\text{H}_3\text{O}^+]/\text{pH} = -\log[\text{H}^+]/[\text{H}_3\text{O}^+] = 10^{-\text{pH}}$ ✓
 (b) Substitute 1,61 in $\text{pH} = -\log[\text{H}_3\text{O}^+]$ ✓
 (c) Calculate $n(\text{HCl})_{\text{unused}}$ using $c = \frac{n}{V}$ ✓
 (d) Calculate $n(\text{HCl})_{\text{initial}}$ using $c = \frac{n}{V}$ ✓
 (e) Calculate $n(\text{HCl})_{\text{used}} = n(\text{HCl})_{\text{initial}} - n(\text{HCl})_{\text{unused}}$ ✓
 (f) Using ratio 1:2 with **USED** HCl from of (e) to calculate $n(\text{CaCO}_3)$ ✓
 (g) Substitute 100 AND $n(\text{CaCO}_3)$ from (f) in $n = \frac{m}{M}$ ✓
 (h) Mass of impurity = $m_{\text{sample}} - m(\text{CaCO}_3)$ ✓
 (i) Final answer: 0,25 g ✓ (Range: 0,2 g to 0,3 g)

Nasienkriteria:

- (a) Enige formule: $\text{pH} = -\log[\text{H}_3\text{O}^+]/\text{pH} = -\log[\text{H}^+]/[\text{H}_3\text{O}^+] = 10^{-\text{pH}}$ ✓
 (b) Vervang 1,61 in $\text{pH} = -\log[\text{H}_3\text{O}^+]$ ✓
 (c) Bereken $n(\text{HCl})_{\text{ongebruik}}$ using $c = \frac{n}{V}$ ✓
 (d) Bereken $n(\text{HCl})_{\text{begin}}$ using $c = \frac{n}{V}$ ✓
 (e) Bereken $n(\text{HCl})_{\text{gebruik}} = n(\text{HCl})_{\text{begin}} - n(\text{HCl})_{\text{ongebruik}}$ ✓
 (f) Gebruik ratio 1:2 van HCl **GEBRUIK** van (e) om $n(\text{CaCO}_3)$ te bereken ✓
 (g) Vervang 100 EN $n(\text{CaCO}_3)$ van (f) in $n = \frac{m}{M}$ ✓
 (h) Massa of onsuiverheid = $m_{\text{monster}} - m(\text{CaCO}_3)$ ✓
 (i) Finale antwoord: 0,25 g ✓ (Gebied: 0,2 g tot 0,3 g)

$\text{pH} = -\log[\text{H}_3\text{O}^+] \quad \checkmark \text{ (a)}$
 $\checkmark 1,61 = -\log[\text{H}_3\text{O}^+]$
 $[\text{H}_3\text{O}^+] = 10^{-1,61}$
 $= 2,45 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3} \text{ (0,0245)}$
 $n(\text{HCl})_{\text{unused}} = n(\text{H}_3\text{O}^+) = cV$
 $= (2,45 \times 10^{-2})(0,2) \quad \checkmark \text{ (c)}$
 $= 4,9 \times 10^{-3} \text{ mol (0,0049)}$
 $n(\text{HCl})_{\text{initial}} = cV$
 $= (0,15)(0,2) \quad \checkmark \text{ (d)}$
 $= 3 \times 10^{-2} \text{ mol (0,03)}$
 $n(\text{HCl})_{\text{used}} = 3 \times 10^{-2} - 4,9 \times 10^{-3} \quad \checkmark \text{ (e)}$
 $= 2,51 \times 10^{-2} \text{ mol (0,0251)}$
 Reaction ratio $n\text{CaCO}_3 : n\text{HCl} = 1:2$
 $n(\text{CaCO}_3) = \frac{1}{2}(2,51 \times 10^{-2}) \quad \checkmark \text{ (f)}$
 $= 1,25 \times 10^{-2} \text{ mol}$
 $n(\text{CaCO}_3) = \frac{m}{M}$
 $1,25 \times 10^{-2} = \frac{m}{100} \quad \checkmark \text{ (g)}$
 $m(\text{CaCO}_3) = 1,25 \text{ g}$
 $m \text{ of impurity in the sample} = 1,5 - 1,25 \quad \checkmark \text{ (h)}$
 $= 0,25 \text{ g} \quad \checkmark \text{ (i)}$

(9)
[22]

QUESTION 8/VRAAG 8

8.1 H^+/H_3O^+ ions/hydrogen ions/hydronium ions/oxonium ions ✓
Waterstofione/hidroniumione/oksoniumione (1)

8.2 0,77 V ✓ (1)

8.3 A ✓ (1)

8.4 H_2 is a stronger reducing agent ✓ than $Fe^{2+}/Fe(II)$ ions ✓ and will reduce $Fe^{3+}/Fe(III)$ ions ✓ (to $Fe^{2+}/Fe(II)$ ions). 

H_2 is 'n sterker reduseermiddel as $Fe^{2+}/Fe(II)$ -ione en sal $Fe^{3+}/Fe(III)$ -ione reduseer (na $Fe^{2+}/Fe(II)$ -ione).

OR/OF

Fe^{2+} -ion is a weaker reducing agent ✓ than H_2 ✓ and therefore $Fe^{3+}/Fe(III)$ ions (to $Fe^{2+}/Fe(II)$ ions) will be reduced. ✓ 

Fe^{2+} -ioon is 'n swakker reduseermiddel as H_2 en sal $Fe^{3+}/Fe(III)$ -ione reduseer (na $Fe^{2+}/Fe(II)$ -ione). (3)

8.5

8.5.1 Pt/Platinum ✓ (1)

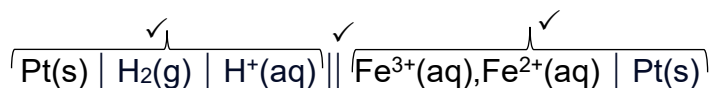
8.5.2 $H_2 \rightarrow 2H^+ + 2e^-$ ✓✓

NOTE/NOTA:

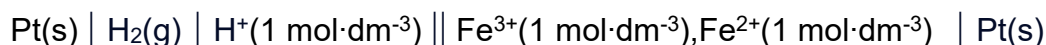
- $2H^+ + 2e^- \leftarrow H_2$ ($\frac{2}{2}$)
 $H_2 \rightleftharpoons 2H^+ + 2e^-$ ($\frac{1}{2}$)
 $2H^+ + 2e^- \rightleftharpoons H_2$ ($\frac{0}{2}$)
 $2H^+ + 2e^- \rightarrow H_2$ ($\frac{0}{2}$)
- Ignore if charge omitted on electron. / Ignoreer indien lading weggelaat op elektron.
- If charge (+) omitted on H^+ / Indien lading (+) weggelaat op H^+ :
 Example/Voorbeeld: $H_2 \rightarrow 2H + 2e^-$ Max/Maks: $\frac{1}{2}$

(2)

8.5.3



OR/OF



ACCEPT/AANVAAR:



(3)

8.6 The reaction reaches equilibrium/no charges/electrons flow. ✓

Die reaksie bereik ewewig/geen ladings/elektrone vloei.

(1)

[13]

QUESTION 9/VRAAG 9**9.1 ANY ONE:**

- The (chemical) process in which electrical energy is converted to chemical energy. ✓✓ **(2 or 0)**
- The use of electrical energy to produce a chemical change.
- Decomposition of an ionic compound by means of electrical energy.
- The process during which an electric current passes through a solution/ionic liquid/molten ionic compound.

ENIGE EEN:

- Die (chemiese) proses waarin elektriese energie omgeskakel word na chemiese energie. ✓✓ **(2 of 0)**
- Die gebruik van elektriese energie om 'n chemiese verandering teweeg te bring.
- Ontbinding van 'n ioniese verbinding met behulp van elektriese energie.
- Die proses waardeur 'n elektriese stroom deur 'n oplossing/ioniese vloeistof/gesmelte ioniese verbinding beweeg.

(2)**9.2 $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ ✓✓****NOTE/NOTA:**

- $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \leftarrow 2\text{H}_2\text{O}(\text{l}) + 2\text{e}^-$ ($\frac{2}{2}$)
 - $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ ($\frac{1}{2}$)
 - $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightleftharpoons 2\text{H}_2\text{O}(\text{l}) + 2\text{e}^-$ ($\frac{0}{2}$)
 - $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \leftarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ ($\frac{0}{2}$)
 - Ignore if charge omitted on electron. // Ignoreer indien lading weggelaat op elektron.
 - If charge (-) omitted on OH^- // Indien lading (-) weggelaat op OH^-
- Example/Voorbeeld: $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}(\text{aq})$ ✓
- Max./Maks: $\frac{1}{2}$
- Ignore phases // Ignoreer fases

(2)**9.3 X to Y/X tot Y ✓****(1)**

9.4

Marking criteria: (a) Substitute 300×10^{-3} and 24 dm^3 into $n = \frac{V}{V_m} \checkmark$ (b) Using ratio 1:2 to calculate $n(e^-)$ ✓ (c) Substitute $6,02 \times 10^{23} \text{ mol}^{-1}$ in $n = \frac{N}{N_A} \checkmark$ (d) Final correct answer: $1,505 \times 10^{22}$ electrons ✓ Range: $1,505 \times 10^{22}$ to $2,41 \times 10^{22}$ electrons	Nasienkriteria: (a) Vervang 300×10^{-3} en 24 dm^3 in $n = \frac{V}{V_m} \checkmark$ (b) Gebruik verhouding 1:2 om $n(e^-)$ te bereken ✓ (c) Vervang $6,02 \times 10^{23} \text{ mol}^{-1}$ in $n = \frac{N}{N_A} \checkmark$ (d) Finale korrekte antwoord: $1,505 \times 10^{22}$ elektrone ✓ Gebied: $1,505 \times 10^{22}$ tot $2,41 \times 10^{22}$ elektrone
OPTION 1/OPSIE 1: $n(\text{Cl}_2) = \frac{V}{V_m}$ $= \frac{300 \times 10^{-3}}{24} \checkmark \text{ (a)}$ $= 0,0125 \text{ mol (0,01)}$ ↓ $n(e^-) = 2n(\text{Cl}_2)$ $= 2(0,0125) \checkmark \text{ (b)}$ $= 0,025 \text{ mol}$ ↓ $n(e^-) = \frac{N}{N_A}$ ↓ $0,025 = \frac{N}{6,02 \times 10^{23}} \checkmark \text{ (c)}$ $N = 1,505 \times 10^{22} \text{ (electrons)} \checkmark \text{ (d)}$	OPTION 2/OPSIE 2: $n(\text{Cl}_2) = \frac{V}{V_m}$ $= \frac{300 \times 10^{-3}}{24} \checkmark \text{ (a)}$ $= 0,0125 \text{ mol (0,01)}$ ↓ $n(\text{Cl}_2) = \frac{N}{N_A}$ $0,0125 = \frac{N}{6,02 \times 10^{23}} \checkmark \text{ (c)}$ $N = 7,525 \times 10^{21} (\text{Cl}_2)$ ↓ $N(e^-) = 2n(\text{Cl}_2)$ $= 2(7,525 \times 10^{21}) \checkmark \text{ (b)}$ $= 1,505 \times 10^{22} \text{ (electrons)} \checkmark \text{ (d)}$

(4)
[9]
150

TOTAL/TOTAAL: